

Distributed Model Predictive Control for Consensus of Constrained Heterogeneous Linear Systems

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Abstract—We consider the problem of steering a multi-agent system to consensus in their outputs. The agents' dynamics are assumed to be heterogeneous, linear, discrete-time and subject to local convex state and input constraints. We present a sequential distributed model predictive control algorithm that asymptotically steers the agents to consensus in their outputs. In their respective model predictive control problems, the agents minimise the distance of a local target output to those of their neighbours while simultaneously tracking the corresponding target steady-state and input pair. We only require the exchange of these target outputs in the scheme whereas the current state and entire predicted trajectories are not shared.

I. INTRODUCTION

Control of a multi-agent system is an important problem for many large-scale applications and it has drawn significant attention in recent years. One important aspect is the consensus problem, in which the goal is to achieve agreement in a certain quantity (output) of the multi-agent system, e.g. bringing a swarm of drones to the same altitude or steering a group of vehicles to a rendezvous. An introduction to consensus for multi-agent systems and an overview of literature is, for example, given in [1]. Most of the existing control methods for establishing consensus do not guarantee satisfaction of constraints on the system dynamics. In this paper, we achieve this by using distributed model predictive control.

Model predictive control (MPC) is a powerful modern optimisation-based control concept for systems subject to constraints. The basic idea of MPC is to repeatedly solve a finite-horizon optimal control problem and then to only apply the first part of the obtained open-loop optimal input sequence, thus introducing feedback. An introduction to MPC is given, for example, in [2]. When considering multi-agent systems, a distributed implementation of MPC is desirable, since this avoids a central coordination and computation unit. This means that each agent locally solves an optimal control problem while communication and cooperation between the agents are needed to accomplish a common objective. A survey of the literature on distributed MPC is provided, for example, in [3], [4] or [5].

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Naturally, consensus via distributed MPC received much attention in recent years. Distributed MPC schemes for multi-agent systems with double integrator dynamics are developed, for example, in [6] and [7]. Consensus for homogeneous linear systems is considered in [8]. While input constraints are usually considered, none of the above can ensure constraints on the state of the agents. In [9], contractive state constraints are imposed in the MPC problem to guarantee consensus for agents with double integrator dynamics subject to input constraints. Consensus via distributed MPC for a multi-agent system with homogeneous, linear dynamics is presented in [10], where constraints on the increment of the state are imposed. Therein, a consensus trajectory is first computed by a consensus protocol and subsequently used in a tracking MPC formulation. In [11] a central optimisation problem is solved to compute an optimal consensus point for general linear systems, which is then used as a setpoint in an MPC formulation. A general framework for cooperative control of dynamically decoupled systems subject to state and input constraints is presented in [12], which can also be used to solve the consensus problem, but requires the sharing of entire predicted state trajectories with neighbours.

In this paper, we propose a distributed MPC scheme for consensus which overcomes the above mentioned limitations: First, it is suitable for general linear, heterogeneous systems. Second, it can guarantee satisfaction of (decoupled) state and input constraints. Third, it only requires the sharing of a target output in each time instant. We achieve this by combining a tracking part and a consensus part in the objective function of the MPC formulation. Each agent tracks a local target steady-state and input pair corresponding to a target output whose distance to the target outputs of its neighbours is simultaneously minimised. This combination of tracking a target and updating it in one MPC problem is introduced in [13], where MPC for tracking piecewise constant references for linear systems with convex constraints is presented. The approach in [13] is extended to a distributed setting in [14] and further in [15] for tracking of piecewise constant references by dynamically coupled linear systems. The basic idea in this approach is the addition of an artificial reference to the decision variables, whose difference from the external reference is penalised in the MPC scheme. In our scheme, instead of tracking an external reference, the agents minimise the distance of their target output to the target outputs of their neighbours. This avoids sharing of initial states or entire predicted trajectories.

Notation. The $n \times n$ -identity matrix is denoted by I_n . We write $A \succ 0$ ($A \succcurlyeq 0$) for a symmetric, positive (semi)definite

matrix $A = A^\top \in \mathbb{R}^{n \times n}$. The maximum and the minimum eigenvalue of a symmetric matrix $A = A^\top \in \mathbb{R}^{n \times n}$ are denoted by $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$. A block diagonal matrix with matrices A_1 to A_m on the diagonal is denoted by $\text{diag}(A_1, \dots, A_m)$. The operator $\otimes$ is the standard Kronecker product. A^+ denotes the Moore-Penrose inverse of a matrix A , see [16]. We denote the Euclidean norm of a vector x by $\|x\|$ and the maximum norm by $\|x\|_\infty$. Let $\omega \in \mathbb{R}$ and $\mathcal{A} \subseteq \mathbb{R}^n$, then $\omega\mathcal{A} = \{\omega a \in \mathbb{R}^n \mid a \in \mathcal{A}\}$. For the Q -weighted (semi)norm with $Q \succ 0$ ($Q \succcurlyeq 0$) we write $\|x\|_Q^2 = x^\top Q x$. The point-to-set distance of a vector $x \in \mathbb{R}^n$ to a set $\mathcal{A} \subseteq \mathbb{R}^n$ is defined as $|x|_{\mathcal{A}} = \inf_{a \in \mathcal{A}} \|x - a\|$.

This paper is arranged as follows. We state the problem in Section II before presenting the main result, the MPC scheme, in Section III. We split this section into three parts, starting with a preliminary tracking problem, extending it then to the final distributed MPC problem and finally stating the algorithm together with an analysis of its properties. In Section IV we illustrate the scheme with a numerical example. Lastly, we give a short summary and an outlook on possible future work in Section V.

II. PROBLEM STATEMENT

We consider a collection of $m \in \mathbb{N}$ agents with linear, discrete-time dynamics

$$x^i(t+1) = A_i x^i(t) + B_i u^i(t), \quad x^i(0) = x_0^i, \quad (1a)$$

$$y^i(t) = C_i x^i(t), \quad (1b)$$

where $x^i(t) \in \mathbb{R}^{n_i}$, $u^i(t) \in \mathbb{R}^{q_i}$ and $y^i(t) \in \mathbb{R}^p$ denote the state, input, and output at time step $t \in \mathbb{N}_0$, x_0^i is the initial condition and the system matrices are $A_i \in \mathbb{R}^{n_i \times n_i}$, $B_i \in \mathbb{R}^{n_i \times q_i}$ and $C_i \in \mathbb{R}^{p \times n_i}$. Note that the agents are heterogeneous; they have the same output dimension but may have different dynamics. The state and input of agent i are constrained to $x^i(t) \in \mathbb{X}_i$ and $u^i(t) \in \mathbb{U}_i$, where $\mathbb{X}_i$ and $\mathbb{U}_i$ are compact and convex polytopes that contain the origin in their interior. We make the following assumption on the dynamics of each agent.

Assumption 1: The pair (A_i, B_i) with A_i, B_i from (1) is stabilisable for all $i \in \mathcal{V}$.

We will make use of the stacked vector form where we stack the vectors of the agents' local variables vertically, denoted by the same letter with the superscript i omitted, e.g. $x(t) = [x^1(t)^\top \dots x^m(t)^\top]^\top \in \mathbb{R}^n$, $n = \sum_{i=1}^m n_i$. Then, with the states, inputs and outputs written this way, the corresponding overall system is given as

$$x(t+1) = \mathcal{A}x(t) + \mathcal{B}u(t), \quad x(0) = x_0, \quad (2a)$$

$$y(t) = \mathcal{C}x(t) \quad (2b)$$

with system matrices $\mathcal{A} = \text{diag}(A_1, \dots, A_m) \in \mathbb{R}^{n \times n}$, $\mathcal{B} = \text{diag}(B_1, \dots, B_m) \in \mathbb{R}^{n \times q}$, $q = \sum_{i=1}^m q_i$, and $\mathcal{C} = \text{diag}(C_1, \dots, C_m) \in \mathbb{R}^{pm \times n}$. The state and input constraints of the overall system are given by the Cartesian product of the local constraints, i.e. $x(t) \in \mathbb{X}_1 \times \dots \times \mathbb{X}_m$ and $u(t) \in \mathbb{U}_1 \times \dots \times \mathbb{U}_m$. Note that the agents' dynamics are completely decoupled and coupling is only introduced by the

later specified control scheme to achieve the common control objective.

The communication topology of the multi-agent system is given by a connected, undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with the set of nodes, i.e. agents, $\mathcal{V} = \{1, \dots, m\}$ and edges $\mathcal{E}$. Agents may bidirectionally share information with each other if they are connected by an edge. The induced set of neighbours of agent i is given as $\mathcal{N}_i = \{j \in \mathcal{V} \mid (i, j) \in \mathcal{E}\}$. The symmetric graph Laplacian is defined as $L = [l_{ij}] \in \mathbb{R}^{m \times m}$ where

$$l_{ij} = \begin{cases} -1, & j \in \mathcal{N}_i, \\ |\mathcal{N}_i|, & j = i, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

For an undirected and connected graph, the Laplacian has non-negative real eigenvalues [1] which we order as follows: $0 = \lambda_{\min}(L) < \lambda_2(L) \leq \dots \leq \lambda_{\max}(L) \leq 2\Delta$ where $\Delta = \max_i |\mathcal{N}_i|$ is the maximum degree of the graph. For a thorough discussion of algebraic graph theory, we refer the reader to [17]. Because the agents' outputs are not necessarily scalar, it proves useful to define the enlarged graph Laplacian $L_p = L \otimes I_p \in \mathbb{R}^{mp \times mp}$. Note that L_p has the same eigenvalues as L .

Our goal is to achieve consensus in the agents' outputs while satisfying each agent's state and input constraints at all times. We accomplish this by using sequential distributed MPC which is proposed in the next section.

III. DISTRIBUTED MODEL PREDICTIVE CONTROL SCHEME

We propose an MPC scheme that features an optimisation problem in which the objective function comprises a tracking part and a consensus part. The latter moves a local target output closer to the target outputs of the neighbours, first resulting in consensus of all target outputs. The tracking part ensures that each agent tracks a local target steady-state and input pair corresponding to the target output. This ensures convergence to consensus in the actual outputs of the agents. For exposition, we first introduce a decoupled tracking MPC formulation and then extend it to the final distributed MPC scheme by adding a suitable cost coupling on the target outputs of neighbouring agents.

A. Preliminary Tracking Problem

In the following, we first consider a decoupled tracking problem where each agent tracks a local admissible target steady-state and input pair $(x_s^i(t), u_s^i(t)) \in \mathcal{S}_i^\omega$. The set $\mathcal{S}_i^\omega$ will be defined below together with an admissible invariant set for tracking, which is introduced in [13]. For clarity, we repeat a similar introduction here. Since (A_i, B_i) is stabilisable by Assumption 1, there exists a matrix K_i such that $A_i + B_i K_i$ is Schur stable. Suppose (1) is controlled by

$$u_{K_i}^i(t) = K_i(x^i(t) - x_s^i(t)) + u_s^i(t). \quad (4)$$

Considering the extended state $z^i(t) = [x^i(t)^\top x_s^i(t)^\top u_s^i(t)^\top]^\top$, we can write the corresponding

closed loop as $z^i(t+1) = A_{z,i}z^i(t)$ with

$$A_{z,i} = \begin{bmatrix} A_i + B_i K_i & -B_i K_i & B_i \\ 0 & I_{n_i} & 0 \\ 0 & 0 & I_{q_i} \end{bmatrix}.$$

The admissible invariant set for tracking is then $\mathcal{O}_i^\omega = \{z^i \mid A_{z,i}^t z^i \in Z_\omega, \forall t \in \mathbb{N}_0\}$ where $Z_\omega = \{(x^i, x_s^i, u_s^i) \mid x^i \in \mathbb{X}_i, K_i(x^i - x_s^i) + u_s^i \in \mathbb{U}_i, (x_s^i, u_s^i) \in \omega \mathbb{X}_i \times \omega \mathbb{U}_i, x_s^i = A_i x_s^i + B_i u_s^i\}$ with $\omega \in (0, 1)$. This set is a convex polytope and can be finitely determined, i.e. defined by a finite number of constraints, for any $\omega \in (0, 1)$, but not necessarily for $\omega = 1$ [18]. For $\omega \rightarrow 1$, the maximum admissible set for tracking is approximated (cf. [13], [18]). Finally, $\mathcal{S}_i^\omega$ are all steady-state and input pairs which can be stabilised by (4) while satisfying the constraints, i.e. $\mathcal{S}_i^\omega = \{(x_s^i, u_s^i) \mid \exists x^i, (x^i, x_s^i, u_s^i) \in \mathcal{O}_i^\omega\}$.

For the purpose of tracking, we define the stage cost $\ell_i(x^i, u^i, x_s^i, u_s^i) = \|x^i - x_s^i\|_{Q_i}^2 + \|u^i - u_s^i\|_{R_i}^2$ with $0 \prec Q_i \in \mathbb{R}^{n_i \times n_i}$ and $0 \prec R_i \in \mathbb{R}^{q_i \times q_i}$. The tracking objective function employed in the predictive control problem below is $J_i^{\text{tr}}(x^i(\cdot|t), u^i(\cdot|t), x_s^i(t), u_s^i(t)) = \sum_{k=0}^{N-1} \ell_i(x^i(k|t), u^i(k|t), x_s^i(t), u_s^i(t)) + V_i^f(x^i(N|t), x_s^i(t))$ with the predicted state trajectory $x^i(\cdot|t) = (x^i(0|t), x^i(1|t), \dots, x^i(N|t))$, the input sequence $u^i(\cdot|t) = (u^i(0|t), \dots, u^i(N-1|t))$, the horizon $N \in \mathbb{N}$ and later specified terminal cost $V_i^f(x^i(N|t), x_s^i(t))$.

Then, the preliminary tracking problem of agent i at time t given the current state $x^i(t)$ and an admissible target steady-state and input pair $(x_s^i(t), u_s^i(t)) \in \mathcal{S}_i^\omega$ is as follows

$$W_i(x^i(t), x_s^i(t), u_s^i(t)) = \min_{u^i(\cdot|t)} J_i^{\text{tr}}(x^i(\cdot|t), u^i(\cdot|t), x_s^i(t), u_s^i(t)) \quad (5a)$$

$$\text{s. t. } x^i(0|t) = x^i(t), \quad (5b)$$

$$x^i(k+1|t) = A_i x^i(k|t) + B_i u^i(k|t), \quad \forall k = 0, 1, \dots, N-1, \quad (5c)$$

$$(x^i(k|t), u^i(k|t)) \in \mathbb{X}_i \times \mathbb{U}_i, \quad \forall k = 0, 1, \dots, N-1, \quad (5d)$$

$$(x^i(N|t), x_s^i(t), u_s^i(t)) \in \mathcal{O}_i^\omega. \quad (5e)$$

The constraints (5b)-(5d) are standard in MPC and are further discussed, for example, in [2]. Note that (5) does not include any information of neighbours. We now define the terminal cost and state an auxiliary result.

Lemma 1: Let Assumption (1) hold. Then there exist a matrix $K_i \in \mathbb{R}^{m_i \times n_i}$ and a matrix $0 \prec P \in \mathbb{R}^{n_i \times n_i}$ such that the following matrix inequality is satisfied

$$(A_i + B_i K_i)^\top P_i (A_i + B_i K_i) - P_i \leq -(Q_i + K_i^\top R_i K_i). \quad (6)$$

Moreover, for any $(x^i(t), x_s^i(t), u_s^i(t)) \in \mathcal{O}_i^\omega$:

$$V_i^f(x_{K_i}^i(t+1), x_s^i(t)) \quad (7a)$$

$$\leq V_i^f(x^i(t), x_s^i(t)) - \ell_i(x^i(t), u_{K_i}^i(t), x_s^i(t), u_s^i(t)), \quad (7b)$$

$$x_{K_i}^i(t+1) \in \mathbb{X}_i, \quad u_{K_i}^i(t) \in \mathbb{U}_i,$$

with $u_{K_i}^i(t)$ as in (4), $x_{K_i}^i(t+1) = A_i x^i(t) + B_i u_{K_i}^i(t)$ and $V_i^f(x^i(t), x_s^i(t)) = \|x^i(t) - x_s^i(t)\|_{P_i}^2$. Furthermore, there exist constants $\beta_i, \eta_i > 0$, such that for any $(x_s^i(t), u_s^i(t)) \in \mathcal{S}_i^\omega$ and any $x^i(t)$ with $\|x^i(t) - x_s^i(t)\|_{Q_i} \leq \eta_i$, the preliminary tracking problem (5) is feasible and

$$W_i(x^i(t), x_s^i(t), u_s^i(t)) \leq \beta_i \|x^i(t) - x_s^i(t)\|_{Q_i}^2. \quad (8)$$

Proof: Since (A_i, B_i) is stabilisable, a matrix K_i exists such that $A_i + B_i K_i$ is Schur stable and the associated Lyapunov equation (6) holds with $P_i \succ 0$ for any $Q_i + K_i^\top R_i K_i \succ 0$. Satisfaction of (7) is standard in MPC and we refer the reader to [2, Chapter 2]. Furthermore, due to the compactness of $\mathbb{X}_i$ and $\mathbb{U}_i$ and the fact that $\omega < 1$, there exists a sufficiently small constant $\alpha_i > 0$ such that for any $(x_s^i(t), u_s^i(t)) \in \mathcal{S}_i^\omega$, $V_i^f(x^i(t), x_s^i(t)) \leq \alpha_i$ implies $(x^i(t), x_s^i(t), u_s^i(t)) \in \mathcal{O}_i^\omega$. Consider $\beta_i = \lambda_{\max}(P_i)/\lambda_{\min}(Q_i)$ and $\eta_i = \sqrt{\alpha_i/\beta_i}$. Then, the bound $\|x^i(t) - x_s^i(t)\|_{Q_i} \leq \eta_i$ implies $V_i^f(x^i(t), x_s^i(t)) \leq \alpha_i$ and thus $(x^i(t), x_s^i(t), u_s^i(t)) \in \mathcal{O}_i^\omega$. Within the admissible set for tracking $\mathcal{O}_i^\omega$, the terminal control law $u_{K_i}^i(t)$ in (4) is a feasible candidate in (5) and the terminal cost $V_i^f(x^i(t), x_s^i(t))$ is an upper bound on $W_i(x^i(t), x_s^i(t), u_s^i(t))$. This last step is also standard in MPC and we refer the reader to [2, Chapter 2], in particular to Proposition 2.18. ■

With this quadratic terminal cost, the preliminary tracking problem is a quadratic program. We are now ready to introduce the coupled optimisation problem that is actually solved by each agent to achieve output consensus.

B. Coupled Optimisation Problem

Our goal is to reach consensus in the output of each agent, i.e. all outputs should eventually be the same. For this reason, we extend the preliminary tracking problem (5) by adding the following term to the objective function

$$V_i^c(y_s^i(t), y_s^{\mathcal{N}_i}(t)) = \frac{1}{2} \sum_{j \in \mathcal{N}_i} \|y_s^i(t) - y_s^j(t)\|^2 \quad (9)$$

where $y_s^i(t)$ is agent i 's target output and $y_s^{\mathcal{N}_i}(t)$ denotes the collected target outputs of agent i 's neighbours. Thus, the pairwise distance between target outputs of neighbouring agents is penalised. In order to allow them to cooperatively agree on the eventual consensus point, we add the target output to the decision variables.

For a given target output y_s^i , a corresponding target steady-state and input pair is defined as a solution of the following system of linear equations

$$\underbrace{\begin{bmatrix} A_i - I_{n_i} & B_i \\ C_i & 0 \end{bmatrix}}_{= \mathcal{T}_i \in \mathbb{R}^{(n_i+p) \times (n_i+q_i)}} \begin{bmatrix} x_s^i \\ u_s^i \end{bmatrix} = \begin{bmatrix} 0 \\ y_s^i \end{bmatrix}. \quad (10)$$

Since there may exist more than one solution, we will always pick the minimum norm solution, which is given by

$$\begin{bmatrix} x_s^i(t) \\ u_s^i(t) \end{bmatrix} = \mathcal{T}_i^+ \begin{bmatrix} 0 \\ y_s^i(t) \end{bmatrix} \quad (11)$$

(cf. [19]). This leads to the set of target outputs $\Omega_i^s = \{y_s^i \in \mathbb{R}^p \mid \exists (x_s^i, u_s^i) \in \mathcal{S}_i^\omega, \mathcal{T}_i \begin{bmatrix} x_s^i \\ u_s^i \end{bmatrix} = \begin{bmatrix} 0 \\ y_s^i \end{bmatrix}, \begin{bmatrix} x_s^i \\ u_s^i \end{bmatrix} = \mathcal{T}_i^+ \begin{bmatrix} 0 \\ y_s^i \end{bmatrix}\}$

of agent i , i.e. target outputs for which the corresponding minimum norm solution of (10) exists and is admissible. Moreover, because we later use a convex combination of neighbouring target outputs to generate a new candidate target output, we further restrict the target output of agent i to the set of admissible target outputs $\mathcal{Y}^s = \cap_{k \in \mathcal{V}} \Omega_k^s$. That is, for each neighbour of agent i with target output $y_s^i(t)$ there also exists a corresponding admissible target steady-state and input pair. Thus, the agents' output sets Ω_i^s need to be shared globally.

The previously introduced tracking cost and (9) are combined into the extended objective function $J_i(x^i(\cdot|t), u^i(\cdot|t), x_s^i(t), u_s^i(t), y_s^i(t), y_s^{\mathcal{N}_i}(t)) = J_i^{\text{tr}}(x^i(\cdot|t), u^i(\cdot|t), x_s^i(t), u_s^i(t)) + 2V_i^c(y_s^i(t), y_s^{\mathcal{N}_i}(t))$. The factor two accounts for the fact that in addition to agent i 's own cost, also the costs of its neighbours that are affected by its decisions are considered. This makes the scheme cooperative as explained in [12]. The resulting optimisation problem of agent i at time step t is an extension of (5) by introduction of cost coupling between the agents through (9) and further constraints. In addition, the local target output is added to the decision variables. Given the current state $x^i(t)$ and the candidate target outputs of the neighbours collected in $\bar{y}_s^{\mathcal{N}_i}(t)$, agent i solves

$$\min_{u^i(\cdot|t), y_s^i(t)} J_i(x^i(\cdot|t), u^i(\cdot|t), x_s^i(t), u_s^i(t), y_s^i(t), \bar{y}_s^{\mathcal{N}_i}(t)) \quad (12a)$$

s. t. (5b)-(5e), (11),

$$y_s^i(t) \in \mathcal{Y}^s. \quad (12b)$$

The target steady-state and input pair $(x_s^i(t), u_s^i(t))$ corresponding to the target output $y_s^i(t)$ is given by (11). We denote the optimising input sequence by $u^{i,*}(\cdot|t)$ and the optimal target input by $y_s^{i,*}(t)$ with the corresponding optimal target steady-state and input pair $(x_s^{i,*}(t), u_s^{i,*}(t))$. Note that (12) is again a quadratic program. In the next section, the scheme that defines the interaction between the agents is proposed and the resulting closed loop is analysed.

C. Distributed MPC Algorithm

It is helpful to first introduce some inequalities and constants. Since the constraints are compact, there exists a constant $y_{\max} > 0$ such that $\|\sum_{j \in \mathcal{N}_i} y_s^i - y_s^j\|^2 \leq y_{\max}$ for all $y_s^i, y_s^j \in \mathcal{Y}^s$. We also need a relationship between target outputs and corresponding target steady-states, which is established in the following lemma.

Lemma 2: For any $y_s^i, \hat{y}_s^i \in \mathcal{Y}^s$ with corresponding $x_s^i, \hat{x}_s^i$ given by (11) there exists a constant $\bar{l}_i \geq 0$ such that

$$\|x_s^i - \hat{x}_s^i\|^2 \leq \bar{l}_i \|y_s^i - \hat{y}_s^i\|^2 \quad (13)$$

Proof: We have $\|x_s^i - \hat{x}_s^i\| \leq \left\| \begin{bmatrix} x_s^i - \hat{x}_s^i \\ u_s^i - \hat{u}_s^i \end{bmatrix} \right\| \stackrel{(11)}{\leq} \|\mathcal{T}_i^+\| \|y_s^i - \hat{y}_s^i\|$ where $\|\mathcal{T}_i^+\|$ is the spectral norm of $\mathcal{T}_i^+$. Squaring both sides yields (13) with $\bar{l}_i = \|\mathcal{T}_i^+\|^2$. ■

Furthermore, we introduce for each agent i the constant step size $\hat{\epsilon}_i = \arg \max_{\epsilon_i} c(\epsilon_i)$ subject to $\epsilon_i \in (0, \min(\Delta^{-1}, \frac{\eta_i}{2}(\lambda_{\max}(Q_i)y_{\max}\bar{l}_i)^{-\frac{1}{2}})]$ where $c(\epsilon_i) =$

$\epsilon_i(2 - \epsilon_i(\lambda_{\max}(L) + 2\beta_i\lambda_{\max}(Q_i)\bar{l}_i))$ and $c(\hat{\epsilon}_i) > 0$. Moreover, $\gamma_i = \min(c_i(\hat{\epsilon}_i)(4\beta_i^2)^{-1}, \eta_i^2(4\beta_i y_{\max})^{-1}, \eta_i^2 y_{\max}^{-1})$ is a positive constant.

In order to achieve the cooperative goal of output consensus, we propose the following algorithm, which is sequential and based on Algorithm 2 in [12]. We assume that before the following algorithm is run, each agent has obtained knowledge of the necessary constants $\lambda_{\max}(L)$ and Δ as well as of the sets Ω_k^s for all $k \in \mathcal{V}$. Thus, the corresponding information has to be exchanged beforehand and each agent has computed its admissible set for tracking offline.

Algorithm 1: 1) At time $t = 0$, each agent finds a feasible candidate input trajectory $\hat{u}^i(\cdot|0)$ and candidate target output $\hat{y}_s^i(0)$ that satisfy (5b)-(5e), and (12b), assuming arbitrary candidate target outputs of its neighbours. Then, each agent transmits its candidate target output $\hat{y}_s^i(0)$ to its neighbours. Go to Step 3.

2) At time instant $t + 1$, each agent i checks if

$$\|x^i(t) - x_s^{i,*}(t)\|_{Q_i} < \gamma_i \left\| \sum_{j \in \mathcal{N}_i} y_s^{j,*}(t) - y_s^{i,*}(t) \right\|^2. \quad (14)$$

a) If the condition is met, take as the new candidate target output

$$\hat{y}_s^i(t+1) = y_s^{i,*}(t) + \hat{\epsilon}_i \sum_{j \in \mathcal{N}_i} y_s^{j,*}(t) - y_s^{i,*}(t). \quad (15)$$

b) If the condition is not met, take as the new candidate target output the previously optimal one, i.e. $\hat{y}_s^i(t+1) = y_s^{i,*}(t)$.

Afterwards, each agent sends $\hat{y}_s^i(t+1)$ to its neighbours.

3) Sequentially, for $i = 1, \dots, m$: Agent i solves problem (12) obtaining a solution $u^{i,*}(\cdot|t)$ and $y_s^{i,*}(t)$. Then, $y_s^{i,*}(t)$ is sent to the agent's neighbours which replace $\hat{y}_s^i(t)$ with $y_s^{i,*}(t)$.

4) Each agent applies $\kappa_{\text{MPC}}^i(t) = u^{i,*}(0|t)$. Go to Step 2.

Note that in this algorithm only the target outputs have to be communicated online with the neighbours. Condition (14) checks whether a target output update or continuing to track the previously optimal target output leads to an overall cost decrease. As a result of Lemma 1, if (14) is met, the step size can be chosen sufficiently small, i.e. $\hat{\epsilon}_i \leq \frac{\eta_i}{2}(\lambda_{\max}(Q_i)y_{\max}\bar{l}_i)^{-\frac{1}{2}}$, such that the target output update (15) is feasible. Note that (15) implements a simple consensus protocol for the target output update (cf. [1]). In the case that (14) does not hold, we take advantage of the fact that the previously optimal target output is again feasible. The restriction of the target outputs to $\mathcal{Y}^s$ allows the use of a convex combination of target outputs. The set of states for which (12) is feasible for all agents is defined as $\mathcal{X}_N = \{x(t) \in \mathbb{R}^n \mid \exists u^{i,*}(\cdot|t) \in \mathbb{R}^{N \times q_i}, y_s^{i,*}(t) \in \mathbb{R}^p \text{ solving (12)} \forall i \in \mathcal{V}\}$.

Before we state the main result which establishes that the agents' outputs converge to consensus and a corresponding consensus set $\mathcal{X}_s^c$ of states is exponentially stable, it is helpful to state some auxiliary inequalities, define the closed-loop system and introduce a Lyapunov function. The consensus

set is defined as $\mathcal{X}_s^c = \{x_s^c \in \mathbb{R}^n \mid \exists y_s^{1,c} = \dots = y_s^{m,c} \in \mathcal{Y}^s, \exists u_s^{i,c}, \begin{bmatrix} x_s^{i,c} \\ u_s^{i,c} \end{bmatrix} = \mathcal{T}_i^+ \begin{bmatrix} 0 \\ y_s^{i,c} \end{bmatrix} \forall i \in \mathcal{V}\}$, i.e. all steady-states of the overall system whose components are (together with an input) the minimum norm solution to an admissible target output that is the same for all agents.

We will make use of the fact that for any $y_s(t)$

$$\|y_s(t)\|_{L_p}^2 \geq \lambda_2(L)\|y_s(t) - P_{L_p}y_s(t)\|^2 \quad (16)$$

with the orthogonal projection $P_{L_p} \in \mathbb{R}^{mp \times mp}$ onto the kernel of L_p (cf. Theorem 3 in [1]). Note that if $y_s(t) \in \mathcal{Y}^s \times \dots \times \mathcal{Y}^s \subset \mathbb{R}^{pm}$, then also $P_{L_p}y_s(t) \in \mathcal{Y}^s \times \dots \times \mathcal{Y}^s \subset \mathbb{R}^{pm}$, since P_{L_p} is doubly stochastic and $\mathcal{Y}^s$ is convex.

The overall closed-loop system is given by

$$x(t+1) = \mathcal{A}x(t) + \mathcal{B}\kappa_{\text{MPC}}(t), \quad x(0) = x_0, \quad (17a)$$

$$y(t) = \mathcal{C}x(t). \quad (17b)$$

We propose a candidate Lyapunov function by inserting the optimal solution of (12) into the objective function and summing up over all agents, that is $V(x(t)) = \sum_{i \in \mathcal{V}} J_i(x^{i,*}(\cdot|t), u^{i,*}(\cdot|t), y_s^{i,*}(t), y_s^{\mathcal{N}_i,*}(t))$. Due to the sequential nature of Algorithm 1, as stated in [12], the upper bound

$$\begin{aligned} V(x(t)) &\leq \sum_{i \in \mathcal{V}} J_i^{\text{tr}}(\hat{x}^i(\cdot|t), \hat{u}^i(\cdot|t), \hat{x}_s^i(t), \hat{u}_s^i(t)) \\ &\quad + \sum_{i \in \mathcal{V}} V_i^c(\hat{y}_s^i(t), \hat{y}_s^{\mathcal{N}_i}(t)) \end{aligned} \quad (18)$$

holds with a candidate target output $\hat{y}_s^i(t)$ and candidate input sequence $\hat{u}^i(\cdot|t)$.

We first establish a lower bound on the Lyapunov function. Note that we say some x_s corresponds to a y_s if x_s^i is (part of) the minimum norm solution of (10) given y_s^i for all $i \in \mathcal{V}$.

Lemma 3: We have $V(x(t)) \geq c_1|x(t)|_{\mathcal{X}_s^c}^2$ for all $x(t) \in \mathcal{X}_N$ with a constant $c_1 > 0$.

Proof: We have

$$\begin{aligned} V(x(t)) &\geq \sum_{i \in \mathcal{V}} \|x^i(t) - x_s^{i,*}(t)\|_{Q_i}^2 + \|y_s^*(t)\|_{L_p}^2 \\ &\stackrel{(16),(13)}{\geq} \sum_{i \in \mathcal{V}} \lambda_{\min}(Q_i) \|x^i(t) - x_s^{i,*}(t)\|^2 \\ &\quad + \sum_{i \in \mathcal{V}} \frac{\lambda_2(L)}{\bar{l}_i} \|x_s^{i,*}(t) - x_s^{i,c}(t)\|^2 \\ &\geq c_1 \|x(t) - x_s^c(t)\|^2 \geq c_1 \|x(t) - \tilde{x}_s^c(t)\|^2 = c_1 |x(t)|_{\mathcal{X}_s^c}^2, \end{aligned}$$

with $c_1 = \frac{1}{2} \min_{i \in \mathcal{V}} (\lambda_{\min}(Q_i), \lambda_2(L)\bar{l}_i^{-1})$, some target steady-state $x_s^c(t)$ corresponding to $y_s^c(t) = P_{L_p}y_s^*(t) \in \ker L_p$ and $\tilde{x}_s^c(t) = \arg \min_{x_s \in \mathcal{X}_s^c} \|x(t) - x_s\|$. Note that for the penultimate inequality, we used the fact that $(a+b)^2 \leq 2a^2 + 2b^2$ for all $a, b \in \mathbb{R}$. ■

Next, we prove that an upper bound on the Lyapunov function exists.

Lemma 4: We have $V(x(t)) \leq c_u|x(t)|_{\mathcal{X}_s^c}^2$ for all $x(t) \in \mathcal{X}_N$ with a constant $c_u > 0$.

Proof: Consider the case where $\|x(t) - \tilde{x}_s^c(t)\|_Q^2 \leq \eta^2$ with $Q = \text{diag}(Q_1, \dots, Q_m)$, $\eta = \min_{i \in \mathcal{V}} (\eta_i)$ and $\tilde{x}_s^c(t) =$

$\arg \min_{x_s \in \mathcal{X}_s^c} \|x(t) - x_s\|$. Then, with Lemma 1, for all $i \in \mathcal{V}$, the target output $\hat{y}_s^{i,c}(t) = \tilde{y}_s^{i,c}(t)$ together with a feasible input sequence $\tilde{u}_s^{i,c}(\cdot|t)$ that minimises the tracking cost part are a feasible candidate in (12), and

$$\begin{aligned} V(x(t)) &\stackrel{(18)}{\leq} \sum_{i \in \mathcal{V}} W_i(x^i(t), \tilde{x}_s^{i,c}(t)) \stackrel{(8)}{\leq} \sum_{i \in \mathcal{V}} \beta_i \|x^i(t) - \tilde{x}_s^{i,c}(t)\|_{Q_i}^2 \\ &\leq \beta_{\lambda_{\max}(Q)} \|x(t) - \tilde{x}_s^c(t)\|^2 = \beta_{\lambda_{\max}(Q)} |x(t)|_{\mathcal{X}_s^c}^2 \end{aligned}$$

with $\beta = \max_i \beta_i$. This upper bound combined with $V(\cdot)$ being upper bounded on the compact set $\mathcal{X}_N$ implies the claimed upper bound (cf. [2, Proposition 2.16]). ■

We are now ready to present the main result which establishes that for the closed-loop system (17) the consensus set $\mathcal{X}_s^c$ is exponentially stable and all agents' outputs converge to the same value, achieving consensus in the agents' outputs.

Theorem 1: Let Assumption (1) hold. Assume that all agents find a feasible candidate for (12) in Step 1 of Algorithm 1. Then, (12) in Step 3 of Algorithm 1 is recursively feasible for all agents and the set $\mathcal{X}_s^c$ is exponentially stable for the closed-loop system (17) with region of attraction $\mathcal{X}_N$. Hence, each output converges to the same value, i.e. $\lim_{t \rightarrow \infty} y^i(t) = \lim_{t \rightarrow \infty} y^j(t)$ for all $i, j \in \mathcal{V}$.

For the sake of brevity, we only sketch the proof. If there is consensus in the optimal target outputs, the corresponding steady-states are stabilised by the tracking part. If not, then at some time step, the tracking cost of the currently tracked optimal target output and corresponding steady-state and input pair becomes sufficiently small. Then, the target output may be incrementally moved such that coupling cost is reduced more than the tracking cost increases. This leads to a decrease in the Lyapunov function in each time step. Note that reusing the previously optimal target output and appending the terminal control law (4) to the previously optimal input sequence is always feasible. In Algorithm 1, the condition (14) is sufficient for a sufficiently small tracking cost such that the update (15) may incrementally move the target output. This eventually leads to consensus in the outputs of the agents while a corresponding steady-state is stabilised.

IV. EXAMPLE

The following example illustrates the presented distributed MPC scheme for a heterogeneous multi-agent system with linear dynamics subject to polytopic constraints. We consider $m = 3$ agents of the form (1) with

$$\begin{aligned} A_1 &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0.2 & 0 & 0 & -0.1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0.9 & 0 & 1 & 0 \\ 0 & 1.1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ B_1 &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \\ C_1 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad C_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \end{aligned}$$

to demonstrate the applicability of the proposed scheme to agents with different state and input dimensions. The goal is to converge to consensus in the outputs, i.e. in the first two state dimensions of the agents. The agents are connected by a line graph, i.e. $\mathcal{N}_1 = \{2\}$, $\mathcal{N}_2 = \{1, 3\}$ and $\mathcal{N}_3 = \{2\}$. Fig. 1

shows that output consensus is achieved. The constraints on the first two state dimensions are depicted in Fig. 1, and the constraints on the last two state dimensions of agents 2 and 3 are given by $\|x_k^i(t)\| \leq 2$ for $i \in \{2, 3\}$ and $k \in \{3, 4\}$. All inputs are constrained by $\|u^i(t)\|_\infty \leq 2$ for $i \in \{1, 2, 3\}$. We compute the admissible sets for tracking with $\omega = 0.99$, the resulting set of admissible target outputs is $\mathcal{Y}^s = \{y^s \in \mathbb{R}^2 \mid y_1^s = 0, y_2^s \in [2.475, -0.990]\}$. The outputs of all agents after 50 time steps are equal to $y(50) = [0.000 \ 2.042]^\top$. We

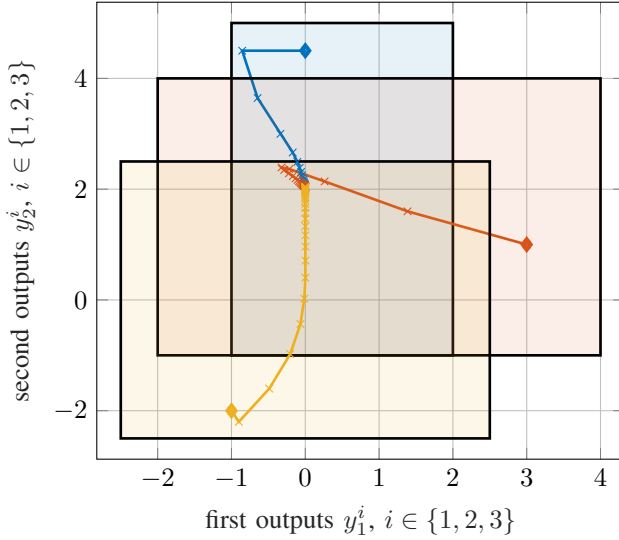


Fig. 1. Evolution of the outputs of all agents. Different time steps are marked with an x, the initial outputs are marked with a diamond. From top to bottom: agent 1, agent 2 and agent 3. Each agent starts in its own constraint set which is shaded.

used the toolbox presented in [20] for the design and plotting of the polytopes.

V. CONCLUSION

We propose a distributed MPC scheme for agents with general linear, heterogeneous dynamics subject to convex state and input constraints that only requires sharing of a target output in each time instant and leads to an online optimisation-based consensus in the agents' outputs. In the MPC problem, each agent tracks a steady-state and input pair that corresponds to a local target output, which is an additional decision variable. In addition, the distance to the target outputs of neighbouring agents is penalised. We show that a set that corresponds to consensus in the outputs of the agents is exponentially stable under application of the proposed MPC scheme. Each agent computes an admissible set for tracking offline which induces a set of target outputs. These sets have to be shared globally and the intersection defines the set of admissible target outputs. This facilitates the use of a simple consensus protocol to generate candidate target outputs in the MPC problems.

In future work, we will explore the possibility of using a projected consensus law with the goal of removing the sharing of target output sets. Furthermore, because the scheme only requires the sharing of a target output in each time

instant and neither the actual output nor entire predicted trajectories need to be communicated to neighbours, one could conjecture that the proposed scheme is suitable for output consensus under privacy concerns. We will more thoroughly study this question in future work.

REFERENCES

- [1] R. Olfati-Saber, J. A. Fax, and R. M. Murray. Consensus and cooperation in networked multi-agent systems. *Proceedings of the IEEE*, 95(1):215–233, 2007.
- [2] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl. *Model predictive control: Theory, computation, and design*. Nob Hill Publishing, Santa Barbara, California, USA, 2nd edition, 2017.
- [3] R. Scattolini. Architectures for distributed and hierarchical model predictive control – a review. *Journal of Process Control*, 19(5):723–731, 2009.
- [4] P. D. Christofides, R. Scattolini, Muñoz de la Peña, D., and J. Liu. Distributed model predictive control: A tutorial review and future research directions. *Computers & Chemical Engineering*, 51:21–41, 2013.
- [5] M. A. Müller and F. Allgöwer. Economic and distributed model predictive control: Recent developments in optimization-based control. *SICE Journal of Control, Measurement, and System Integration*, 10(2):39–52, 2017.
- [6] Z. Cheng, H.-T. Zhang, M.-C. Fan, and G. Chen. Distributed consensus of multi-agent systems with input constraints: A model predictive control approach. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 62(3):825–834, 2015.
- [7] L. Zhou and S. Li. Distributed model predictive control for consensus of sampled-data multi-agent systems with double-integrator dynamics. *IET Control Theory & Applications*, 9(12):1774–1780, 2015.
- [8] H. Li and W. Yan. Receding horizon control based consensus scheme in general linear multi-agent systems. *Automatica*, 56:12–18, 2015.
- [9] G. Ferrari-Trecate, L. Galbusera, M.P.E. Marciandi, and R. Scattolini. Model predictive control schemes for consensus in multi-agent systems with single- and double-integrator dynamics. *IEEE Transactions on Automatic Control*, 54(11):2560–2572, 2009.
- [10] J. Lee, J.-S. Kim, H. Song, and H. Shim. A constrained consensus problem using MPC. *International Journal of Control, Automation and Systems*, 9(5):952, 2011.
- [11] B. Johansson, A. Speranzon, M. Johansson, and K. H. Johansson. Distributed model predictive consensus. In *Proc. International Symposium on Mathematical Theory of Networks and Systems*, pages 2438–2444, Kyoto, Japan, 2006.
- [12] M. A. Müller, M. Reble, and F. Allgöwer. Cooperative control of dynamically decoupled systems via distributed model predictive control. *International Journal of Robust and Nonlinear Control*, 22(12):1376–1397, 2012.
- [13] D. Limon, I. Alvarado, T. Alamo, and E. F. Camacho. MPC for tracking piecewise constant references for constrained linear systems. *Automatica*, 44(9):2382–2387, 2008.
- [14] A. Ferramosca, D. Limon, J. B. Rawlings, and E. F. Camacho. Cooperative distributed MPC for tracking. *IFAC Proceedings Volumes*, 44(1):1584–1589, 2011.
- [15] C. Conte, M. N. Zeilinger, M. Morari, and C. N. Jones. Cooperative distributed tracking MPC for constrained linear systems: Theory and synthesis. In *Proceedings of the 52nd IEEE Conference on Decision and Control*, pages 3812–3817, Florence, Italy, 2013.
- [16] R. Penrose. A generalized inverse for matrices. *Mathematical Proceedings of the Cambridge Philosophical Society*, 51(3):406–413, 1955.
- [17] C. Godsil and G. Royle. *Algebraic graph theory*, volume 207. Springer New York, New York, USA, 2001.
- [18] E. G. Gilbert and K. T. Tan. Linear systems with state and control constraints: the theory and application of maximal output admissible sets. *IEEE Transactions on Automatic Control*, 36(9):1008–1020, 1991.
- [19] M. James. The generalised inverse. *The Mathematical Gazette*, 62(420):109, 1978.
- [20] M. Herceg, M. Kvasnica, C.N. Jones, and M. Morari. Multi-Parametric Toolbox 3.0. In *Proc. of the European Control Conference*, pages 502–510, Zürich, Switzerland, July 17–19 2013. <http://control.ee.ethz.ch/~mpt>.